Data Release ReadMe

Automatic Weather Station, Surface Height Change, Snow Temperature, Albedo, and Black Carbon/Dust Observations from Gulkana Glacier

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This document serves as the documentation for the data release containing field observations taken on Gulkana Glacier in summer, 2024.

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   Complete weather station dataset from the on-ice station ("gulkanaB_AWS_2024.csv")

   Preprocessed, simulation-ready dataset of weather station data ("single_summer_sim_data.csv")

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1. Automatic Weather Station

gulkanaB_AWS_2024.csv

This file contains the quality-controlled meteorological data from the on-ice weather station (Table 1). Timestamps are in UTC, and the local time zone is AKDT, UTC-08:00.

The station was installed at Gulkana Site B in April on fresh snow. It was allowed to lower with the depleting snowpack until it rested on ice. In August, it was found to be on smooth, flat ice prior to uninstallation. See photos for visual context.

Table 1: Automatic weather station sensors

Sensor	Manufacturer	Variable measured	Variable name	Units	Aggregation type
HygroVUE	Campbell Scientific	Air temperature	AirTC_Avg	°C	Average
		Relative humidity	RH	%	Sample
BaroVue 10	Campbell Scientific	Surface pressure	BV_BP_Avg	hPa	Average
WindMonitor-HD	R.M. Young	Wind speed	WS_ms_Avg	$m\ s^{-1}$	Average
		Gust speed	WS_ms_Max	$m\ s^{-1}$	Maximum
CNR4	Kipp & Zonen	Incoming shortwave radiation	SWUpper_Avg	$W\ m^{-2}$	Average
		SWUpper variability	SWUpper_Std	$W\ m^{-2}$	Standard deviation
		Outgoing shortwave radiation	SWLower_Avg	$W\ m^{-2}$	Average
		SWLower variability	SWLower_Std	$W\ m^{-2}$	Standard deviation

Instrumentation details

A CR1000X datalogger was used to communicate with and record data from the sensors.

The HygroVue was installed at 2 m ±10 cm above the surface in a MetSpec RAD10 10plate radiation shield.

The WindMonitor-HD was installed at 2.3 m ±10 cm above the surface. Upon uninstallation in August, it was discovered that the rotational piece was not locked into place, negating confidence in any wind direction measurements. This data was thus discarded.

The CNR4 was installed on a crossarm 1.5 m above the surface. It was levelled upon installation and field workers were careful not to disturb the snow directly under the sensor. The station was visited and leveled on June 21st. Upon arrival in August, sensor tilt was measured to be 2°.

The longwave sensors on the CNR4 were improperly wired to the logger, so all longwave data was discarded.

The CNR4 was both heated and ventilated. The program ran the heater/ventilator as follows:

- Heater turned on when the 10-min average air temperature was measured below –2°C
- Heater turned off when the 10-min average air temperature was measured above 0°C
- Ventilator operated continuously throughout the entire period

Power Requirements

Due to the large power requirements to ventilate and heat the CNR-4, two separate battery systems were installed.

Battery #1 was a standard 24 Ah lead-acid battery stored within the main enclosure with the CR1000X datalogger. This battery was charged via 50 W solar panel and discharged to power all the logging sensors. Voltage was monitored by the CR1000X and is reported as VBatt in the data.

Batteries #2 and 3 were 100 Ah ReLion lithium-ion batteries connected in parallel. They were stored within a separate enclosure which was mounted on the vertical member opposite the crossarm with the CNR4 to act as counterweight. These batteries were

charged using a 90 W solar panel and discharged only to the heater/ventilator (Kipp & Zonen CNF4-L).

Quality Control

Air temperature, relative humidity, wind speed, surface pressure, battery voltage were left in their original form without quality control.

Incoming and outgoing shortwave have been filtered for the following conditions:

- Removed values above 1000 W m^{-2}
- Removed values below 0 W m^{-2}
- Removed values where outgoing shortwave exceeded incoming shortwave

single_summer_sim_data.csv

This file includes the preprocessed data used to run the Python Energy Balance model for Snow and Ice (PEBSI) for single-summer simulations. Most of it is equivalent to that in the “GulkanaB_AWS_2024.csv” file but resampled from 10-minute resolution to hourly averages. Variables which are not used in the simulation are removed, including:

- Outgoing shortwave data
- Shortwave radiation standard deviations
- Wind gust speeds

The data is converted to the units expected by PEBSI.

- Air temperature (°C), relative humidity (%), and wind speed (m s^{-1}) are in original units
- Incoming shortwave and longwave radiation are converted from W m^{-2} to J m^{-2}
- Surface pressure is converted from hPa to Pa

Incoming longwave data is taken from the off-ice USGS weather station at 1725 m a.s.l. Please refer to the USGS weather station data release (U.S. Geological Survey Glacier Project, 2019) for more details.

2. Snow temperature

gulkanaB_iButtons_2024.csv

This file contains snow temperature measurements from five iButton temperature sensors (±0.5°C precision). The sensors were painted white such that they absorb minimal radiation when below the snow surface.

The sensors were installed in the wall of a 270-cm-deep snow pit. After sampling the pit for snow density using a 1000 cc wedge, the wedge cuts were used to line up the sensors at evenly-distributed depths (top sensor 10 cm below the initial snow surface, and bottom sensor resting on the ice surface.) The end of an ice ax was used to push the sensors at least 15 cm into the wall. The snow pit was then filled in.

The sensors each had a piece of flagging so they could be found in August resting on the ice surface.

Data Description:

- Columns are labeled by each sensor's initial height above the ice in cm
    - 0 = resting on the ice surface
    - 260 = 10 cm below initial snow surface
- Timestamps are in UTC

3. Surface height change

gulkana{site}_{GNSSIR/stake}_2024.csv

These files contain daily surface height change measurements determined by two methods. For details on the sensors and processing, please see Wells et al. (2024).

Data Description:

- Two types of files are present: “GNSSIR” and “stake”
  - “GNSSIR” is data processed from a GNSS receiver on top of a stake
  - “stake” is data from a camera pointed at the base of the banded ablation stake
- Columns contain:
  - Date
  - CMB: cumulative surface height change in meters relative to 4/18/2024
  - sigma: reported error of GNSS-IR data points in meters

The GNSS systems were installed at six sites, five of which operated for at least part of the summer. A steam drill was used to create holes at each site in which a 10 ft banded ablation stake was installed. The stakes had an insulating wooden dowel on the bottom to prevent heat conduction from the metal pipe causing melting. The stakes were installed with approximately a meter above the surface. GNSS units were placed atop each stake and a time lapse camera was placed at the base pointing at the intersection of stake and snow/ice. The camera provides measurements of surface height change from the passing of bands in each image, while the GNSS-IR uses the signal-to-noise ratio from the sensor to determine the surface height change.

The associated Journal of Glaciology study uses only GNSS-IR for Sites AB, B, BD, and T and stake data for Site D. However, all existing data from the 2024 melt season are provided here, including that unused in the study.

4. Light-absorbing particles

gulkana_LAPs_2024.csv

This file contains black carbon and dust measurements from two sample dates in June and August 2024. Samples were collected using a custom 3D printed cutter measuring 10cm x 10cm x 3cm. Samples were cut to 12 cm bottom depth (four sections) in the side of a shallow snow pit. Samples were kept cold (rush shipped on dry ice) until analysis at the Snow HydRO Lab (University of Utah). Of the August samples, black carbon was only measured in the 0-3 cm samples. For more details on the laboratory methodology, please see Skiles and Painter (2017).

Data Description:

- Columns contain details about each sample:
    - ID: arbitrary integer assigned to each sample
    - sample date: date the sample was collected in the field
    - site: location of the sample
    - sample depth: depth range from the snow surface of the sample
    - sample mass: total mass of the sample in g
    - dust concentration: concentration in mg L-1 determined by gravimetric method
    - black carbon concentration: concentration in ug L-1 determined by Single Particle Soot Photometer

5. Albedo

gulkana{site}_{date}_spectrum.csv

Albedo measurements were recorded using an ASD FieldSpec4 High-Res spectroradiometer (350–2500 nm) and cosine receptor foreoptic at two sites in June 2024. All measurements were collected within 2 hours of solar noon on mostly cloudy days. Two methods were used:

Method 1: “albedo”

The foreoptic sensor is placed on a level mount. Ten upward measurements are taken, then the sensor is turned downward and ten more measurements are taken in succession. In data processing, the upward and downward spectra are averaged and divided to provide a single reflectance spectrum. This type of measurement is done when albedo for a single location in space is of interest.

Method 2: “reflectance transects”

A field-of-view limiter is placed on the foreoptic sensor. The sensor is first held out over a white reference sheet with a nominal albedo of 1. The user then walks around and takes distributed measurements every ~2 meters. In data processing, each reflectance spectra is divided by the average of the 10 white reference spectra. Because lighting conditions change rapidly, white reference spectra are retaken repeatedly every ~10 minutes. This type of measurement is done when spatially distributed albedo is of interest.

Albedo measurements were taken at Gulkana B using Method 1; two measurements on 19 June 2024 and two on 20 June 2024. The indexers 1 and 2 in the data are arbitrary; both measurements were taken within 30 minutes.

A reflectance transect using Method 2 was taken near the terminus of Gulkana on bare ice and debris on 21 June 2024. The data release includes the average between the 2 measurements where the broadband (integrated) albedo exceeds 0.35, which are assumed to see clean ice instead of debris/rocks. The data includes this average spectrum. If the debris data is of interest, please reach out to the authors of this data release.

All files include two columns: wavelength in nm and reflectance (unitless).

gulkanaB_daily_albedo_2024.csv

A daily timeseries of broadband albedo from the automatic weather station. Albedo is calculated as the ratio of measured upward and downward shortwave radiation. The daily albedo is aggregated by taking the average of 10:00 to 14:00 local time measurements.

6. Site locations

File contains the latitude, longitude, and approximate elevation of each site referenced in the data.

7. AWS photos

Three images of the on-ice weather station were taken on 18 April, 20 June, and 20 August 2024 by C. Wilson.

References

Wells, A. et al. (2024) “GNSS reflectometry from low-cost sensors for continuous in situ contemporaneous glacier mass balance and flux divergence”, Journal of Glaciology, 70, p. e5. doi:10.1017/jog.2024.54.

Skiles, S.M. and Painter, T. (2017) “Daily evolution in dust and black carbon content, snow grain size, and snow albedo during snowmelt, Rocky Mountains, Colorado,” Journal of Glaciology, 63(237), pp. 118–132. doi:10.1017/jog.2016.125.

U.S. Geological Survey Glacier Project (2019) High altitude weather station data at USGS Benchmark Glaciers (ver 2.0, December 2024). U.S. Geological Survey data release, doi:10.5066/P9EUXIPE